

Mitochondrial disorders in chagasic cardiomyopathy.

PMID: 15769630 | Indexed in PubMed

The mitochondria play a complex multi-factorial role in the cell. Along with their primary role in energy (ATP) production, mitochondria generate reactive oxygen species (ROS) that directly or indirectly affect several cellular functions. In this article, I review the molecular, structural and functional mitochondrial abnormalities reported in chagasic cardiomyopathy. I highlight current information about the potential etiology and the pathophysiological significance of mitochondrial dysfunction in chagasic cardiomyopathy and provide a brief background of mitochondrial biogenesis and bioenergetic pathways in cardiac growth and development.

Mitochondrial biogenesis and function in Arabidopsis.

PMID: 22303236 | Indexed in PubMed

Mitochondria represent the powerhouse of cells through their synthesis of ATP. However, understanding the role of mitochondria in the growth and development of plants will rely on a much deeper appreciation of the complexity of this organelle. Arabidopsis research has provided clear identification of mitochondrial components, allowed wide-scale analysis of gene expression, and has aided reverse genetic manipulation to test the impact of mitochondrial component loss on plant function. Forward genetics in Arabidopsis has identified mitochondrial involvement in mutations with notable impacts on plant metabolism, growth and development. Here we consider the evidence for components involved in mitochondria biogenesis, metabolism and signalling to the nucleus.

Mitochondrial AAA+ proteases.

PMID: 37945172 | Indexed in PubMed

Mitochondria are multifunctional organelles that play a central role in a wide range of life-sustaining tasks in eukaryotic cells, including adenosine triphosphate (ATP) production, calcium storage and coenzyme generation pathways such as iron-sulfur cluster biosynthesis. The wide range of mitochondrial functions is carried out by a diverse array of proteins comprising approximately 1500 proteins or polypeptides. Degradation of these proteins is mainly performed by four AAA+ proteases localized in mitochondria. These AAA+ proteases play a quality control role in degrading damaged or misfolded proteins and perform various other functions. This chapter describes previously identified roles for these AAA+ proteases that are localized in the mitochondria of animal cells.

The mitochondrial respiratory chain of the secondary green alga *Euglena gracilis* shares many additional subunits with parasitic Trypanosomatidae.

PMID: 24561571 | Indexed in PubMed

The mitochondrion is an essential organelle for the production of cellular ATP in most eukaryotic cells. It

is extensively studied, including in parasitic organisms such as trypanosomes, as a potential therapeutic target. Recently, numerous additional subunits of the respiratory-chain complexes have been described in *Trypanosoma brucei* and *Trypanosoma cruzi*. Since these subunits had apparently no counterparts in other organisms, they were interpreted as potentially associated with the parasitic trypanosome lifestyle. Here we used two complementary approaches to characterise the subunit composition of respiratory complexes in *Euglena gracilis*, a non-parasitic secondary green alga related to trypanosomes. First, we developed a phylogenetic pipeline aimed at mining sequence databases for identifying homologues to known respiratory-complex subunits with high confidence. Second, we used MS/MS proteomics after two-dimensional separation of the respiratory complexes by Blue Native- and SDS-PAGE both to confirm in silico predictions and to identify further additional subunits. Altogether, we identified 41 subunits that are restricted to *E. gracilis*, *T. brucei* and *T. cruzi*, along with 48 classical subunits described in other eukaryotes (i.e. plants, mammals and fungi). This moreover demonstrates that at least half of the subunits recently reported in *T. brucei* and *T. cruzi* are actually not specific to Trypanosomatidae, but extend at least to other Euglenozoa, and that their origin and function are thus not specifically associated with the parasitic lifestyle. Furthermore, preliminary biochemical analyses suggest that some of these additional subunits underlie the peculiarities of the respiratory chain observed in Euglenozoa.

Apoptosis and necrosis in health and disease: role of mitochondria.

PMID: 12722948 | Indexed in PubMed

Mitochondria play an important role in both the life and death of cells. Mitochondria are the powerhouse of the cell, providing over 90% of adenosine triphosphate (ATP) consumed by the cell. Mitochondrial energy production, however, is disrupted in various pathological situations leading to cellular injury. The mechanisms causing the injury are turning out to be more complex than originally expected. For instance, calcium, oxidant chemicals, ischemia/ reperfusion, and a range of other agents promote onset of the mitochondrial permeability transition in mitochondria from liver, heart, and other tissues. Often the consequence of this event is ATP depletion, ion deregulation, mitochondrial and cellular swelling, activation of degradative enzymes, plasma membrane failure, and cell lysis. This is referred to as necrotic cell death. The mitochondrial permeability transition is also involved in apoptotic cell death. In this mode of death, the role of the permeability transition is to release proapoptotic proteins from mitochondria into the cytosol where with the aid of cellular ATP they complete the apoptotic cascade. Therefore, mitochondria contribute to both apoptotic and necrotic death.

Mitochondria and mitochondrial disorders: an overview update.

PMID: 35843711 | Indexed in PubMed

Mitochondria, the cell powerhouse, are membrane-bound organelles present in the cytoplasm of almost all the eukaryotic cells. Their main function is to generate energy in the form of adenosine triphosphate (ATP). In addition, mitochondria store calcium for the cell signaling activities, generate heat, harbor pathways of intermediate metabolism and mediate cell growth and death. Primary mitochondrial diseases (MDs) form a clinically as well as genetically heterogeneous group of inherited disorders that result from the mitochondrial energetic metabolism malfunctions. The lifetime risk of the MDs development is estimated at 1:1470 of newborns, which makes them one of the most recurrent groups of inherited disorders with an important burden for society. MDs are progressive with wide range of

symptoms of variable severity that can emerge congenitally or anytime during the life. MD can be caused by mutations in the mitochondrial DNA (mtDNA) or nuclear DNA genes. Mutations inducing impairment of mitochondrial function have been found in more than 400 genes. Furthermore, more than 1200 nuclear genes, which could play a role in the MDs' genetic etiology, are involved in the mitochondrial activities. However, the knowledge regarding the mechanism of the mitochondrial pathogenicity appears to be most essential for the development of effective patient's treatment suffering from the mitochondrial disease. This is an overview update focused on the mitochondrial biology and the mitochondrial diseases associated genes.

Mitochondria: structure and function.

PMID: 1962047 | Indexed in PubMed

Mitochondria are the main site of ATP synthesis in aerobic cells, using the free energy of the oxidation of metabolic fuels by oxygen. They have a matrix space containing the enzymes of the citrate cycle and beta-oxidation, enclosed by an inner membrane containing the 4 complexes of the electron transport chain, ATP synthase and specific carriers for metabolites. Mitochondria also have a relatively permeable outer membrane and an intermembrane space. ATP synthesis (oxidative phosphorylation) is critically dependent on the structural integrity of the mitochondrion. Electrons from substrate oxidations feed into the electron transport chain at complex I or complex II, and then successively flow to complex III, complex IV and finally to oxygen. Complexes I, III and IV are redox pumps and electron transport causes extrusion of protons from the matrix generating an electrochemical proton gradient (proton motive force) across the inner membrane. Protons return to the matrix 'through' ATP synthase driving the synthesis of ATP. The stoichiometry of proton extrusion and the yield of ATP are still uncertain. Mitochondria have genetic continuity and are inherited maternally. They possess a small amount of DNA which codes for some, but not all, of the subunits of complexes I, III, IV of ATP synthase. mtDNA also codes for mitochondrial ribosomal and messenger RNAs involved in the synthesis of mitochondrially coded subunits. All other mitochondrial peptides are synthesised on cytosolic ribosomes and are imported and targeted to their specific intramitochondrial locations, often after proteolytic removal of leader sequences.

The effects of kahweol, a diterpene present in coffee, on the mitochondria of the human neuroblastoma SH-SY5Y cells exposed to hydrogen peroxide.

PMID: 31306736 | Indexed in PubMed

The oxidative phosphorylation (OXPHOS) system located in the mitochondria is the main source of adenosine triphosphate (ATP) in mammals. The mitochondria are also the main site of reactive oxygen species (ROS) production in those cells. Disruption of the mitochondrial redox biology has been seen in the onset and progression of neurodegenerative diseases. In this regard, we have tested here whether kahweol (KW; C₂₀H₂₆O₃), a diterpene present in coffee, would be able to promote mitochondrial protection in the human neuroblastoma SH-SY5Y cells exposed to hydrogen peroxide (H₂O₂). A pretreatment (for 12 h) with KW (at 10 μ M) decreased the impact of H₂O₂ (at 300 μ M) on the levels of oxidative stress markers in the mitochondrial membranes, as well as reduced the production of ROS by the organelles. KW pretreatment also suppressed the effects of H₂O₂ on the activity of components of

the OXPHOS. The KW-induced mitochondria-related effects were blocked by inhibition of the phosphoinositide 3-kinase/Akt (PI3K/Akt) and p38 mitogen-activated protein kinase (MAPK) signaling pathways. Furthermore, silencing of the transcription factor nuclear factor erythroid 2-related factor 2 (Nrf2) and inhibition of the heme oxygenase-1 (HO-1) enzyme abrogated the KW-induced protective effects on the mitochondria. Therefore, KW promoted mitochondrial protection by the PI3K/Akt and p38 MAPK/Nrf2/HO-1 axis in H₂O₂-challenged SH-SY5Y cells.

TP53-induced glycolysis and apoptosis regulator is indispensable for mitochondria quality control and degradation following damage.

PMID: 29375708 | Indexed in PubMed

Mitochondria have been described as 'the powerhouse of the cell' as the organelle generates the majority of adenosine triphosphate (ATP) in cells to support life. Mitochondria can be damaged due to stress, for example by reactive oxygen species (ROS). TP53-induced glycolysis and apoptosis regulator (TIGAR) serves a role in suppressing ROS damage and may protect mitochondria integrity. In the present study, the localization of TIGAR on mitochondria in 5-8F cells was demonstrated. Furthermore, it was indicated that the knockdown of TIGAR using lentivirus-short hairpin RNA induces the loss of mitochondrial membrane potential and cytochrome c leakage. However, these damaged mitochondria were not degraded in cells, but exhibited an abnormal appearance as indicated by mitochondrial swelling, crista collapse and vacuolization, with physiological dysfunction marked by reduced ATP production. Therefore, TIGAR maybe an indispensable protein for mitochondrial protection and degradation following cellular damage.

Editorial for Special Issue "Plant Mitochondria".

PMID: 30513904 | Indexed in PubMed

The primary function of mitochondria is respiration, where catabolism of substrates is coupled to adenosine triphosphate (ATP) synthesis via oxidative phosphorylation (OxPhos). [...].